Natural Hazards (2025) 121:13447–13473
https://doi.org/10.1007/s11069-025-07326-9

ORIGINAL PAPER

Heat vulnerability index mapping through principal component analysis and equal weight methods: comparing Spatial patterns at a low urban scale and local climate zones in an arid mid-size South American coastal city

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Received: 8 November 2024 / Accepted: 21 April 2025 / Published online: 6 May 2025

Abstract
One of the most evident effects of the cities expansions is the increasing urban temperatures, the so-called urban heat islands, affecting more significantly vulnerable populations. In this study, we examine heat vulnerability in Arica, a medium-sized coastal and desert city in Chile, using the Heat Vulnerability Index (HVI) methodology. We compared the performance of two methodological approaches: Principal Component Analysis (PCA) and Equal Weight (EW), as well as how this vulnerability is distributed among local climate zones. The results show that the general spatial pattern of heat vulnerability indices is moderately heterogeneous in the city. The PCA method grouped all indicators into four independent components that explain 71.7% of the total variance. Specifically, the PCA method overestimated vulnerability in the urban core of the city as well as in Cancha Rayada sector, while the EW method overestimated it in peripheral areas due to inherent differences in how each method weighs and processes spatial data. Regarding vulnerability to heat in the local climatic zones (LCZ) of Arica, both methods agreed that the Lightweight lowrise represents the class with the highest levels of vulnerability, notably in areas such as Cancha Rayada and Pedro Blanquier, the districts most susceptible to extreme heat. This study may help to provide results for future research, as well as to carry on adaptation and mitigation strategies for urban planning in Arica, in a context of implementation of the climate change adaptation and mitigation plans in Chile and the necessary actions at the urban level.

Keywords Arica · Equal weight · Local climate zones · Principal component analysis · Urban heat Islands

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1 Introduction

Climate change is one of the most pressing problems of our time, and one of its most notable effects has been the rise in global temperatures. The Intergovernmental Panel on Climate Change (IPCC) has predicted that this trend, reflected in the increase in heat waves since the end of the 20th century, will continue with increasing frequency and intensity worldwide (IPCC 2022).

This situation represents a direct threat to people's health, especially for the most vulnerable groups, which experience high incidence and mortality related to heat waves (Kenny et al. 2010; Chien et al. 2016). The vulnerability of these groups is usually attributed to sociodemographic conditions, medical problems, and geographical location (Arsad et al. 2022). To address this challenge, climate researchers have developed several heat vulnerability indices, providing a new way to study extreme heat and its possible effects. In this article, a Heat Vulnerability Index (HVI) is defined as a multidimensional tool used in climate research and public health to assess and measure the susceptibility of populations to extreme heat events based on socioeconomic and physical-environmental indicators.

This definition is based on a comprehensive framework composed of the concepts of Exposure, Sensitivity, and Adaptive Capacity, through which many authors have applied their HVIs (Wolf and McGregor 2013; Inostroza et al. 2016; Méndez-Lázaro et al. 2018; Mallen et al. 2019). Characterizing vulnerability from these concepts is challenging due to the dynamic interaction between them, as vulnerability may vary depending on the population and local context (Conlon et al. 2020).

In this regard, maps have become the primary technique for characterizing a Heat Vulnerability Index (HVI) and identifying the most vulnerable areas of a given city. They provide a clear and accessible way to visualize data, communicate vulnerability levels effectively, and prioritize resources for intervention and mitigation. Furthermore, this spatial visualization technique facilitates the implementation of public policies, supporting public health institutions in planning specific population interventions and raising awareness of health risks, making it an indispensable tool for urban planning (Jeong et al. 2024).

When deciding to map heat vulnerability indices, it is crucial to consider the spatial scale of analysis. Different scales can produce different results, so the choice of an appropriate spatial unit depends on several factors, such as the heterogeneity of the urban environment, population distribution, and data availability. In this study, we have chosen to use census blocks and Local Climatic Zones (LCZ) as spatial units. While census blocks are commonly used in several studies, LCZ are not used as frequently. However, in recent years, researchers have begun to consider these units. This is because most studies tend to focus only on the thermal conditions of the environment, overlooking the vulnerability to heat of different human groups and their adaptation capacities, depending on the characteristics of the urban environment provided by the LCZ (Cai et al. 2019).

1.1 Heat vulnerability in coastal desert cities: South American context and local background

When studying heat events in coastal desert cities, the challenge becomes even greater due to the spatiotemporal heterogeneity and the significant influence of synoptic climatic conditions (Yun et al. 2020). In southwestern South America, specifically the Atacama Desert, this heterogeneity has been previously studied (Morales-Salinas et al. 2023). Additionally, this subregion faces scenarios where social inequalities, urban expansion, and inadequate planning expose populations to climate change effects, increasing the risk of floods, landslides, vegetation cover deterioration, and heat-related hazards (IPCC 2022).

Northern Chile, one of the driest regions in the world, is part of this context. It is estimated that summer months could become even hotter, and heat waves could increase their frequency up to tenfold (Feron et al. 2019). Consequently, these scenarios make our study area, the city of Arica, a coastal desert city vulnerable to heat hazards. At the regional level, which includes Arica and other municipalities, the Climate Risk Atlas projections for Chile (ARClim 2021) estimate an increase of 1.6 ºC in the annual average maximum daily temperature and 25.9 days of heat waves by 2050, which could result in 27.2 net deaths due to increased temperatures during the hot season.

When citizens face extreme heat threats, vegetation emerges as a fundamental factor in mitigation. According to the Chilean National Institute of Statistics (Instituto Nacional de Estadística [INE] 2018), Arica has 3.75 m² of vegetation per inhabitant. This figure is notably below the World Health Organization (WHO) and the National Urban Development Council (CDNU) recommendations of 9 m² and 10 m², respectively. However, relying solely on vegetation for heat mitigation in arid cities is complex due to inherent conditions. Thus, other factors more suitable for such environments have been emphasized, such as the selection of urban structures (Gourfi et al. 2022).

This underscores the importance of considering Local Climate Zones (LCZ) in arid environments like Arica. In Chile, there has been recent interest in their application, positioning LCZ as a significant framework for studies on heat and its effects from a spatial perspective. LCZ provide homogeneous units that help guide urban planning strategies, allowing cities and their residents to adapt to specific climatic conditions (Smith et al. 2023). Despite this background, other studies in Arica are conducted using projection methodologies at larger scales (communal or regional level), identifying results related to Arica from a general perspective and ignoring more local factors (Quintana-Talvac et al. 2021). Moreover, desert cities do not occupy a prominent position in the literature on Heat Vulnerability Indices (HVI) (Li et al. 2022).

In consideration of the problems described and to unravel the complexities of heat vulnerability in Arica, this study applies an HVI tailored to the particularities of the city.

1.2 An HVI for Arica

An HVI, as defined previously, aggregates various vulnerability indicators into a single score to identify locations at high risk for heat exposure (Liu et al. 2020). With this purpose in mind, several indicators were selected from official sources to construct the index. For this study, two of the most commonly used methods for creating HVIs were employed: Principal Component Analysis (PCA) and Equal Weight (EW), resulting in two distinct HVIs. In PCA, indicators undergo statistical analysis, receiving weights based on their variability or relative importance within the dataset. Conversely, EW assigns equal importance to each vulnerability indicator, excluding the variability and providing a neutral view that facilitates direct comparison disregarding the variability of the indicators.

Both methods, however, have recognized limitations. EW can produce inconsistencies by not capturing the true influence of each indicator on the final index while PCA assigns weights that may not always accurately reflect the actual importance of each indicator, depending on the dataset (Price et al. 2024). The literature reflects an ongoing debate regarding these methodological differences, particularly in studies of heat vulnerability indices. Jeong et al. (2024) highlight that most studies focus on regional analyses using socioeconomic variables, as is the case with the present work. Considering this, they recommend comparing heat vulnerability using multiple methods rather than relying solely on one, as demonstrated by the works of Liu et al. (2020) and Guo et al. (2019). Based on this premise, it is understood that each method can yield different spatial outcomes, highlighting different vulnerable areas. The comparison between PCA and EW is expected to provide a comprehensive understanding of heat vulnerability, which is particularly useful in urban contexts like Arica, where heat vulnerability mapping is still in its early stages of development.

Therefore, the main objective of this study is to map and compare the results of both methodologies and their spatial patterns to evaluate and characterize heat vulnerability within Arica's urban census boundary, according to the differences detected between PCA and EW. Additionally, the methods were analyzed across the Local Climate Zones (LCZ) to better understand their applicability in the local context. This comparative analysis aims not only to identify areas of high vulnerability but also to understand the differences between the methodological approaches and their suitability at the local level. Ultimately, this analysis is expected to contribute to the design of effective heat mitigation strategies and sustainable urban planning in Arica, leveraging the specific characteristics of local climatic zones to obtain more precise and contextualized results.

2 Materials and methods

2.1 Study area

Arica (18°28'S) is a medium-sized city and commune that also serves as the capital of the Arica and Parinacota Region, located at the northernmost tip of Chile along the Pacific Ocean (Fig. 1). The study area was defined using the urban census boundary provided by the INE, and included two levels of analysis: the census districts (16 in total) and their respective census tracts (70 in total). The communal area of Arica spans 4,799.40 km^2, while the study area encompasses 45.2 km^2 of the city. This study area includes 205,079 inhabitants, representing 92.64% of the commune's population and 90.71% of the entire regional population.

Geographically, Arica features a distinctive arid landscape. According to the Köppen Climate Classification, Arica is classified as BW (Sarricolea et al. 2017). Other sources classify it as BWn, indicating a desert climate with significant cloudiness (Hernández Palma et al. 2014). This cloudiness, often referred to as "camanchaca", is a dense morning fog that is prevalent due to the influence of the cold Humboldt Current.

Climatologically, Arica has an average annual temperature of 18 °C, which never falls below 15 °C during winter and can reach up to 30 °C in summer (DMC 2020). Annual precipitation is exceptionally scant, totaling less than 3 mm (González et al. 2016).

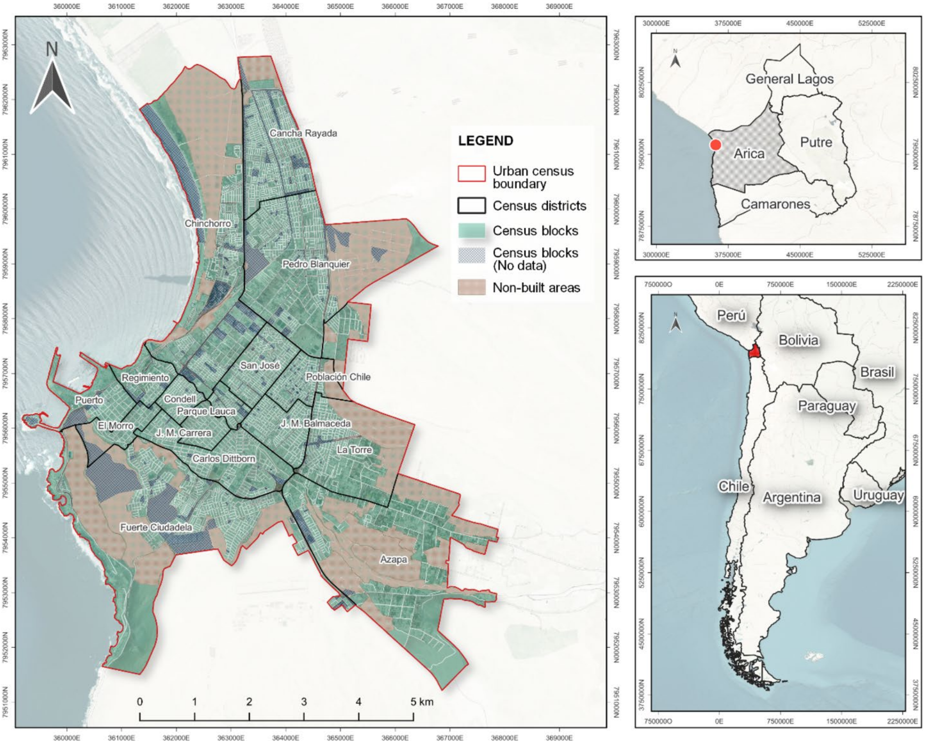

Fig. 1 Location of the study area

2.2 HVI construction and data

The HVIs were developed using spatial analysis techniques, and the spatial patterns of heat vulnerability were mapped for the predefined study area in Arica. As outlined in the introduction, the formulation of these indices is conceptualized as a function or sum of Exposure, Sensitivity, and Adaptive Capacity. Exposure measures the heat intensity or its severity risk at a specific location (Oke 1987). Sensitivity determines the potential impact of heat exposure on individuals. Adaptive Capacity refers to the population's skills or resources to resist adverse heat conditions (Cutter et al. 2008).

The indicators that compose the HVIs and their respective components, as discussed earlier, were selected. Socioeconomic variables were sourced from the INE's 2017 Population and Housing Census and the 2021 Territorial Human Wellbeing Matrix (MBHT for its Spanish acronym), developed by the Center for Territorial Intelligence at Adolfo Ibáñez University (UAI for its Spanish acronym). The following subsection provides a personal but justified description of the selected indicators. This justification is based on bibliographic references where these indicators have been used in other HVIs under the same conceptual framework (Exposure, Sensitivity, and Adaptive Capacity). Also, other types of HVIs and similar works with different indexing methods were considered. Table 1 shows a summary of all the chosen indicators.

Table 1 Summary of data sources and normalization for selected indicators

Dimension	N°	Indicators	Calculation of Variables	Data Source / Year or Period	Normalization according to directionality
Exposure	1	LST	Mean pixel values by census tracts, 30 m	Landsat TM, ETM+, TIRS / 1992–2022	Positive
	2	Population density	Number of people/ha	Population and Housing Census 2017 (INE)	Positive
Sensitivity	1	Elderly population	Number of older adults aged 65 years or more/ha	Population and Housing Census 2017 (INE)	Positive
	2	Infant population	Number of infants 5 years old or younger/ha	Population and Housing Census 2017 (INE)	Positive
	3	Female population	Number of women/ha	Population and Housing Census 2017 (INE)	Positive
	4	Educational level	Zonal mean of the average number of years of education of household heads	Territorial Human Well-being Matrix 2021 (UAI)	Negative
	5	Employability	Zonal mean of the proportion of the labor force with employment	Territorial Human Well-being Matrix 2021 (UAI)	Negative
Adaptive capacity	1	Housing quality	Zonal mean of all materials per household (walls, floors, ceilings)	Territorial Human Well-being Matrix 2021 (UAI)	Positive
	2	Access to water supply	Number of households with access to the public water network per ha	Population and Housing Census 2017 (INE)	Positive
	3	Access to health centers	Zonal mean of public and private health centers per inhabitant	Territorial Human Well-being Matrix 2021 (UAI)	Negative
	4	Urban green areas	Area of local green areas that serve the population	Territorial Human Well-being Matrix 2021 (UAI)	Negative
	5	NDVI	Mean pixel values by census tracts, 30 m	Landsat TM, ETM+, TIRS / 1992–2022	Negative

2.2.1 Exposure indicators

(1) Land Surface Temperature (LST): This indicator identifies areas subjected to higher heat risks, aiding in the assessment of population vulnerability under high temperature conditions (Johnson et al. 2012; Wolf and McGregor 2013; Inostroza et al. 2016; Méndez-Lázaro et al. 2018; Zhang et al. 2018; Bhattacharjee et al. 2019; Zuhra et al. 2019; Raja et al. 2021; Abrar et al. 2022).

(2) Population Density: Highly populated areas may face increased challenges due to heat. It is hypothesized that a high concentration of people may amplify exposure and elevate the demand for services (Wolf and McGregor 2013; Loughnan et al. 2014; Zuhra et al. 2019; Raja et al. 2021; Gaglione et al. 2022).

2.2.2 Sensitivity indicators

(1) Elderly Population: This indicator highlights the proportion of the population typically aged 60 or 65 years and older, who are most susceptible to the effects of extreme heat

and require special protective measures (Reid et al. 2009; Johnson et al. 2012; Wolf and McGregor 2013; Loughnan et al. 2014; Zhu et al. 2014; Bradford et al. 2015; Inostroza et al. 2016; Méndez-Lázaro et al. 2018; Mallen et al. 2018; Conlon et al. 2020; Raja et al. 2021; Abrar et al. 2022; Gaglione et al. 2022).

(2) Infant Population: Represents children aged 5 years or younger, who are the second most vulnerable age group to extreme heat (Loughnan et al. 2014; Zhu et al. 2014; Inostroza et al. 2016; Zhang et al. 2018; Gaglione et al. 2022).

(3) Female Population: Considers the proportion of females, whose vulnerability to heat is associated with specific biological and social factors (Johnson et al. 2012; Romero-Lankao et al. 2012; Raja et al. 2021; Abrar et al. 2022).

(4) Educational Level: Reflects the population's educational attainment, which may influence their understanding of heat risks and ability to take protective measures (Reid et al. 2009; Johnson et al. 2012; Inostroza et al. 2016; Zhang et al. 2018; Zuhra et al. 2019; Gaglione et al. 2022).

(5) Employability: Indicates the economic capacity of individuals to access means of protection against heat (Zhu et al. 2014; Inostroza et al. 2016; Méndez-Lázaro et al. 2018; Raja et al. 2021; Gaglione et al. 2022).

2.2.3 Adaptive capacity indicators

(1) Housing Quality: Measures the quality of housing materials, where lower-quality materials may enhance heat absorption, increasing vulnerability (Romero-Lankao et al. 2012; Inostroza et al. 2016; Bhattacharjee et al. 2019; Zuhra et al. 2019; Samuelson et al. 2020).

(2) Access to Water Supply: Assesses the availability of water from public networks, essential for hydration and cooling during extreme heat conditions (Inostroza et al. 2016; Zuhra et al. 2019; Prosdocimi and Klima 2020; Raja et al. 2021; Abrar et al. 2022).

(3) Access to Health Centers: Highlights proximity to health centers, crucial for providing medical care during heat-related emergencies (Romero-Lankao et al. 2012; Loughnan et al. 2014; Inostroza et al. 2016; Bhattacharjee et al. 2019; Abrar et al. 2022).

(4) Urban Green Areas: Considers the availability of public green spaces such as parks and squares, which can function as cooling shelters during periods of intense heat (Reid et al. 2009; Loughnan et al. 2014; Bradford et al. 2015; Raja et al. 2021; Abrar et al. 2022; Gaglione et al. 2022).

(5) NDVI: Measures the quality and density of vegetation, aiding in heat mitigation by providing shade and reducing local temperatures (Johnson et al. 2012; Inostroza et al. 2016; Bhattacharjee et al. 2019; Zuhra et al. 2019; Prosdocimi and Klima 2020; Abrar et al. 2022).

According to Inostroza et al. (2016), to mitigate spatial bias, socioeconomic variables from the census were normalized per hectare for each census district. Conversely, variables from the MBHT were averaged, reflecting their derivation from other pre-normalized indices. Regarding physical-environmental variables, two were chosen: the Land Surface Temperature (LST) and the Normalized Difference Vegetation Index (NDVI), both acquired from the Landsat 5, 7, and 8 satellites. The methodology for this process is described subsequently.

The following subsections provide a detailed and justified description of the selected indicators. This justification draws on bibliographic sources where these indicators have been utilized in other HVIs and within the same conceptual framework of Exposure, Sensitivity, and Adaptive Capacity. Additionally, this analysis considers other types of HVIs and related studies employing different indexing methods.

2.3 Preprocessing and data normalization

A critical step was to ensure that all indicators consistently reflected heat vulnerability in a unidirectional manner. Specifically, higher values of each indicator should correspond to greater vulnerability, and conversely, lower values should signify reduced vulnerability. Given this necessity, the indicators associated with Educational Level, Employability, Access to Health Centers, Urban Green Areas, and NDVI were adjusted accordingly. Before incorporating these indicators into the HVIs through PCA and EW methods, both positive and negative normalization techniques were applied. This adjustment ensured that all indicators pointed in the same direction, where a value of 1 represents the highest level of vulnerability to heat, and a value of 0 indicates the lowest level (Table 1).

2.4 LST and NDVI

We considered a 30-year study period ranging from 1992 to 2022. We retrieved satellite images from Landsat 5, 7, and 8 captured during the summer months (December to March). The specific scene for the city of Arica is located at Path 002/Row 073. These images were accessed and downloaded from the Earth Explorer website of the United States Geological Survey (USGS) at https://earthexplorer.usgs.gov/. The spatial resolution of these satellite images is 30 m, with thermal bands resampled by the USGS to align with this resolution. It should be noted that some images from the Landsat 7 satellite exhibited the well-documented Scan Line Error. To address this issue and ensure the accuracy and reliability of our analysis, these images were restored using Gap Mask files provided by the USGS.

A crucial part of this procedure involved the meticulous selection of cloud-free images. Although cloud cover percentages are typically calculated for entire images, this approach can misrepresent smaller study areas like ours. To mitigate this, we employed stringent visual criteria to examine each image and applied cloud masks where necessary, leading to the exclusion of some images. Prior to the calculation of LST and NDVI, atmospheric correction using the DOS1 method was applied to enhance the accuracy of the results. All selected images are detailed in Table 2.

The calculation of LST for Landsat 5 TM and Landsat 7 ETM + images involves several steps. Initially, digital numbers must be converted to radiance using the following equation:

$$L\lambda \ = M_L * Q_{cal} + A_L$$

Where $L\lambda$ is the top of atmospheric spectral radiance calculated in watts/(m^2 ∗srad ∗µm), M_L is the band specific multiplicative rescaling factor; Q_{cal} is the digital level captured by the sensor, and A_L is the band specific additive scaling factor.

Once the spectral radiance is obtained, it is converted to brightness temperature (TB) as follows:

$$BT = \frac{K2}{ln(\frac{K1}{L\lambda} + 1)} - 273.15$$

Table 2 Landsat images used in the analysis

Landsat ID	Landsat Satellite Missions	Acquisition Date
LT05_L1TP_002073_19920228_20200914_02_T1	Landsat 5 (TM)	1992/02/28
LT05_L1TP_002073_19970108_20200910_02_T1	Landsat 5 (TM)	1997/01/08
LT05_L1TP_002073_19970124_20200911_02_T1	Landsat 5 (TM)	1997/01/24
LE07_L1TP_002073_20020114_20200917_02_T1	Landsat 7 ETM+	2002/01/14
LE07_L1TP_002073_20020130_20200917_02_T1	Landsat 7 ETM+	2002/01/30
LE07_L1TP_002073_20020319_20211023_02_T1	Landsat 7 ETM+	2002/03/19
LE07_L1TP_002073_20021216_20200916_02_T1	Landsat 7 ETM+	2002/12/16
LE07_L1TP_002073_20070128_20200913_02_T1	Landsat 7 ETM+	2007/01/28
LE07_L1TP_002073_20070301_20200913_02_T1	Landsat 7 ETM+	2007/03/01
LE07_L1TP_002073_20071214_20200913_02_T1	Landsat 7 ETM+	2007/12/14
LE07_L1TP_002073_20120110_20200909_02_T1	Landsat 7 ETM+	2012/01/10
LE07_L1TP_002073_20120126_20200909_02_T1	Landsat 7 ETM+	2012/01/26
LE07_L1TP_002073_20120211_20200909_02_T1	Landsat 7 ETM+	2012/02/11
LE07_L1TP_002073_20120227_20200909_02_T1	Landsat 7 ETM+	2012/02/27
LC08_L1TP_002073_20170131_20200905_02_T1	Landsat 8 (TIRS)	2017/01/31
LC08_L1TP_002073_20170216_20200905_02_T1	Landsat 8 (TIRS)	2017/02/16
LC08_L1TP_002073_20171201_20200902_02_T1	Landsat 8 (TIRS)	2017/12/01
LC08_L1TP_002073_20220113_20220123_02_T1	Landsat 8 (TIRS)	2022/01/13
LC08_L1TP_002073_20220302_20220309_02_T1	Landsat 8 (TIRS)	2022/03/02
LC08_L1TP_002073_20221231_20230110_02_T1	Landsat 8 (TIRS)	2022/12/31

Where *K1* and *K2* are thermal conversion constant varying for both TIR bands, $L\lambda$ is the top of atmospheric spectral radiance, and 273.15 corresponds to the transformation from °K to °C.

For Landsat 8 TIRS images, the same steps are followed, but in the calculation of the spectral radiance, the band 10 correction is subtracted. However, after determining BT, it is necessary to determine the emissivity to finally obtain the LST image.

Before that, the NDVI must be calculated from the Near Infrared (NIR) and Red (RED) bands.

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

This computation reports the quantity and quality of vegetation through the reflection of chlorophyll in the near-infrared spectral region. NDVI produces values between −1 and 1 for each pixel. From this result, the Vegetation Portion (Pv) is calculated:

$$Pv = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}}\right)^2$$

Where *NDVImax* represents the maximum digital levels of the previous product, and *NDVImin* represents the minimum digital levels. From this step, the pixels emissivity (ε) with mixed coverage is calculated as following:

$$\epsilon = 0.004Pv + 0.986$$

Finally, after these previous calculations, the following equation is applied to determine the LST or the emissivity-corrected land surface temperature T_s:

$$T_s = \frac{BT}{1 + \left(\frac{\lambda\, BT}{\rho}\right) ln\epsilon}$$

Where T_s is the LST in °C, BT is at-sensor BT, λ is the wavelength of emitted radiance (11.5 µm for TM and ETM+sensors, 10.8 µm for TIRS sensor), ρ is a constant value of $1.438 \cdot 10^{-2}$ m K based on the Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K), the Planck constant ($6.626 \cdot 10^{-34}$ J s), and the velocity of light ($2.998 \cdot 10^{8}$ m/s), and ε is the calculated emissivity.

Like the LST calculation process, the resulting NDVI images for each date were subjected to the same averaging and correction process. The resulting images were cropped with the urban census boundary, and the zonal statistics tool was applied to obtain the average values of both temperature and vegetation for each census tracts.

2.5 EW and PCA approaches

In the Equal Weighting (EW) approach, all selected indicators were assigned equal importance in the index, operating under the assumption that each indicator equally influences heat vulnerability. In contrast, the Principal Component Analysis (PCA) adopts a multidimensional approach by extracting uncorrelated variables (principal components) that capture the maximum variance within the dataset. For our analysis, a Varimax Rotation was used to enhance interpretability, achieving this by maximizing the variance accounted for by each principal component. This rotation facilitates a simplified index structure while preserving the explained variance.

In the development of the indices using the PCA and EW methods, a critical phase was the comparison of their results. To facilitate this comparison, the results of both indices were standardized using z-scores (Table 3). By adjusting the range from a minimum of -1.25 or less to a maximum of 1.25 or more, the z-values were transformed to span a new range from −3 to 3. This adjustment enabled a meaningful comparison between the two indices, providing a comprehensive assessment of the dynamics of heat vulnerability.

2.6 LCZ classification

Land surface temperature and both heat vulnerability indices were averaged according to the local climatic zones of Arica. The first through local statistics and the second through the

Table 3 Assignment of HVI values to EW and PCA results

Range values	HVI Values	Vulnerability Level
< −1.25	-3	Very Low
−1.25 to −0.75	-2	Low
−0.75 to −0.25	-1	Medium – Low
−0.25 to 0.25	0	Medium
0.25 to 0.75	1	Medium - High
0.75 to 1.25	2	High
> 1.25	3	Very High

union of attributes by location. The layer of the local climatic zones are the same ones used in previous studies (Smith et al. 2021, 2023). The LCZs correspond to building types (10):

- compact highrise (1): Tall multi-storey buildings, dense mix, no trees, mostly paved ground. Materials: concrete, steel, stone, glass.
- compact midrise (2): Mid-rise buildings (3–9 storeys), dense mix, no trees, mostly paved ground. Materials: stone, brick, tiles, concrete.
- compact lowrise (3): Low-rise buildings (1–3 storeys), dense mix, no trees, mostly paved ground. Materials: stone, brick, tiles, concrete.
- open highrise (4): Tall buildings with open layout, permeable ground with low floors. Materials: concrete, steel, stone, glass.
- open midrise (5): Open layout mid-rise buildings, permeable ground with low floors. Materials: concrete, steel, stone, glass.
- open lowrise (6): Open plan low-rise buildings, permeable ground with low floors. Materials: wood, brick, stone, tiles, concrete.
- lightweight lowrise (7): Single-storey buildings, no trees, mostly compacted soil. Lightweight materials: wood, straw, corrugated meta.
- large lowrise (8): Large low-rise buildings (1–3 storeys), no trees, mostly paved ground. Materials: steel, concrete, metal, stone.
- sparsely built (9): Small to medium sized dispersed building, low ground cover.
- heavy industry (10): Low to medium height industrial buildings, mostly paved or hardened ground. Materials: steel, metal, concrete.

Seven land cover types are also described:

- dense trees (A): dense landscape of deciduous and/or evergreen trees, mostly permeable soil, natural forest function or urban park.
- scattered trees (B): Light landscape of scattered trees, mostly permeable soil, natural forest function or urban park.
- bush, scrub (C): Landscape of low shrubs and trees, mostly bare or sandy soil, natural or agricultural function.
- low plants (D): Landscape of grasses or herbaceous plants, no trees, flat ground, agricultural or urban park function.
- bare rock or paved (E): Bare rock or paved landscape, no trees or plants, desert or urban transport function.
- bare soil or sand (F): Bare soil or sand landscape, without trees or plants, desert or agricultural function.
- water (G): Large bodies of water such as seas and lakes or small bodies of water such as rivers and lagoons.

3 Results

3.1 Exploring Arica LST distribution

The calculation of the average daily LST for the period from 1992 to 2022 during the summer months revealed surface temperatures ranging from 18 °C to 36 °C (Fig. 2). Low temperatures, between 18 °C and 26 °C, are predominantly found along the entire coastal edge of the city. However, while similar temperatures occasionally occur inland (e.g., Azapa), they are not characteristic of any specific census district. Intermediate temperatures, ranging from 26 °C to 30 °C, follow a parallel distribution to the low temperatures along the coast

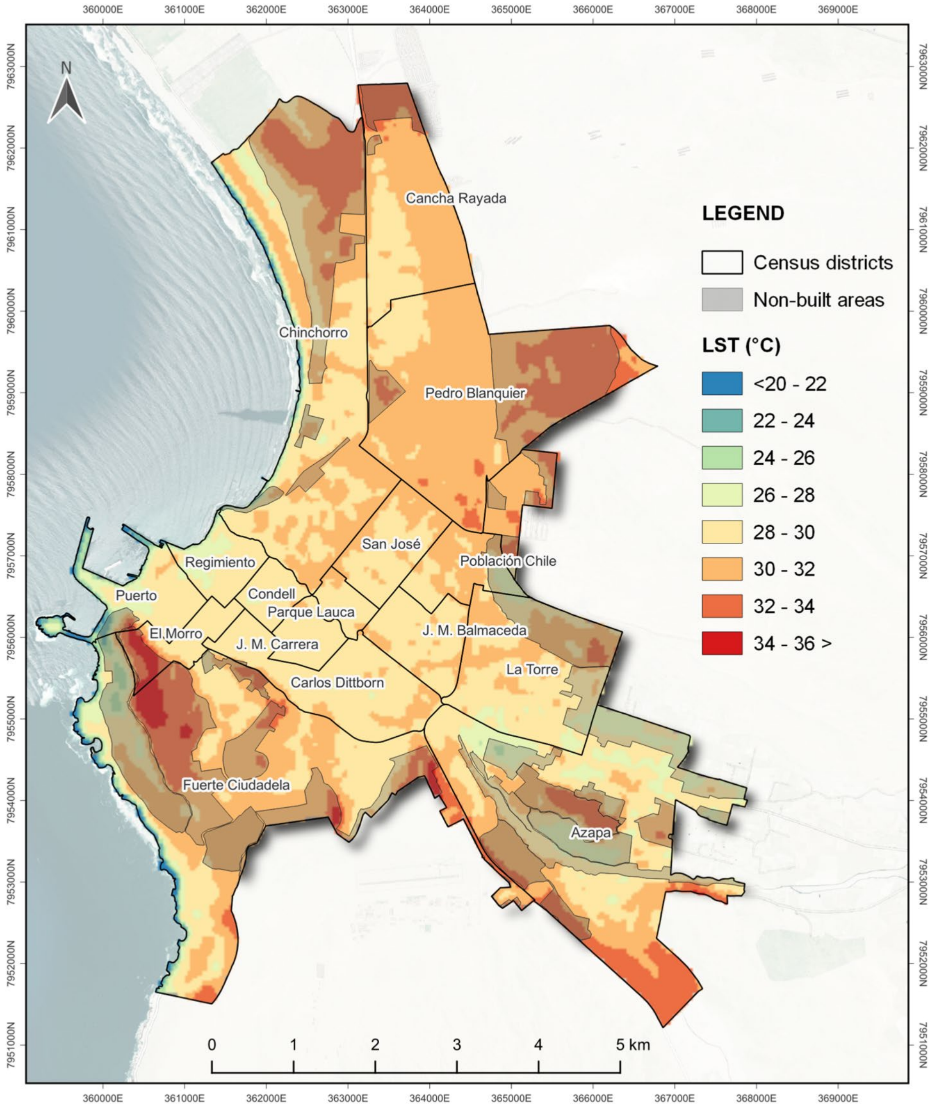

Fig. 2 Daily summer LST in Arica between 1992 and 2022

but extend further into the interior of the city. These temperatures are most pronounced in the central zone of Arica, particularly in predominantly urban census districts such as Puerto, Regimiento, Condell, J.M. Carrera, J.M. Balmaceda, Parque Lauca, San José, and Carlos Dittborn, and they extend to La Torre and to lesser-developed areas such as Azapa, albeit to a lesser degree in the northern and southern parts of the city.

In contrast, the highest temperatures, between 34 °C and 36 °C, are concentrated in areas far from the urban center. In the north, these temperatures span the census districts of Pedro Blanquier (the most intense), Cancha Rayada, and Chinchorro. Similarly, in the southern zone, they encompass the districts of El Morro and Fuerte Ciudadela, featuring a significant thermal band that extends considerably southeast towards the border of the Azapa district.

3.2 Spatial patterns of heat vulnerability in Arica using PCA

Using the PCA method with Varimax Rotation, the 12 indicators employed in this study were grouped into 4 independent components, cumulatively explaining 71.7% of the variance. Specifically, the first component accounts for 31.4% of the variance, the second for 15.4%, the third for 13.3%, and the fourth for 11.6% (Table 4). For that, Component 1 comprises Population density, Infant population, Female population and Access to water supply; Component 2 comprises Elderly population, Access to health cares and Urban green areas; Component 3 comprises Educational level and Employability; and Component 4 comprises LST, Housing quality and NDVI (Fig. 3).

The first component comprises Population Density, Elderly Population, Infant Population, and Female Population. The second component includes Educational Level, Employability, and Housing Quality. The third component is made up of Access to Water Supply, Access to Health Centers, and Urban Green Areas. The fourth component consists of LST and NDVI.

Figure 4 indicates that in the first component, the highest HVI values are observed in the census districts of Cancha Rayada and Pedro Blanquier, where census tracts do not fall below medium vulnerability. Conversely, districts such as Azapa, El Morro, Puerto, and

Table 4 Number of extracted factors, eigenvalues, and percentages of variance explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	% Cumulative	Total	% of Variance	% Cumulative	Total	% of Variance	% Cumulative
1	3.874	32.285	32.285	3.874	32.285	32.285	3.768	31.397	31.397
2	2.202	18.346	50.631	2.202	18.346	50.631	1.846	15.381	46.777
3	1.386	11.550	62.181	1.386	11.550	62.181	1.599	13.324	60.101
4	1.148	9.571	71.752	1.148	9.571	71.752	1.398	11.651	71.752
5	0.985	8.212	79.964						
6	0.618	5.153	85.117						
7	0.546	4.554	89.671						
8	0.462	3.849	93.520						
9	0.354	2.948	96.468						
10	0.323	2.694	99.162						
11	0.082	0.686	99.848						
12	0.018	0.152	100.000						

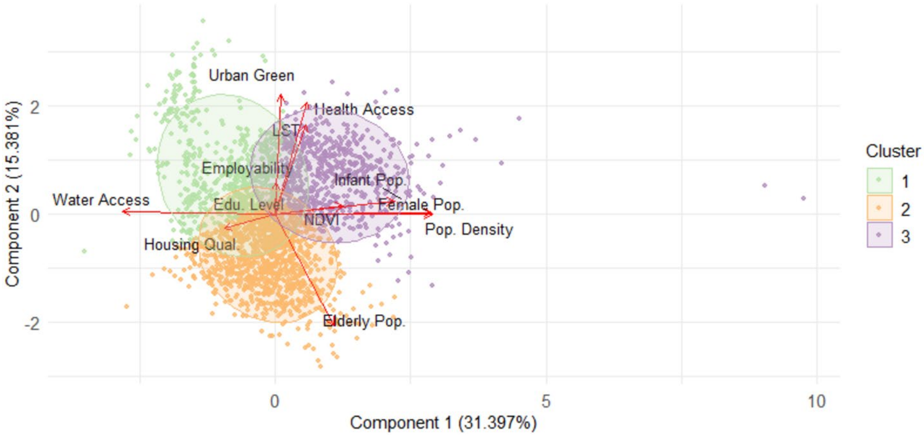

Fig. 3 Contribution of each variable for components 1 and 2

Regimiento predominantly exhibit low values. In contrast, the second component, which includes indicators of Education, Employability, and Housing Quality, shows a notable shift. Here, some of the less vulnerable census tracts from the first component now represent higher levels of vulnerability, while those that were more vulnerable initially maintain their status in this component.

Component 3, characterized by the indicators of Access to Water Supply, Access to Health Centers, and Urban Green Areas, shows that Azapa exhibits uniformly very high vulnerability levels. This is followed by Pedro Blanquier, Fuerte Ciudadela, Cancha Rayada, Chinchorro, and La Torre, where the remaining districts display lower susceptibility to heat. In contrast, in component 4, where LST and NDVI predominate, the Azapa district exhibits very low vulnerability to heat. Meanwhile, districts previously identified as high vulnerability areas in other components continue to oscillate between medium and extremely high vulnerability levels.

3.3 Spatial patterns of heat vulnerability in Arica using EW

Figure 5 illustrates the mapping of indicators that contributed to the calculation of the HVI using the EW method. These indicators were normalized and are represented using natural ranges. As previously mentioned, the LST indicator displays its highest values in the districts of Pedro Blanquier, Cancha Rayada, and Fuerte Ciudadela, followed to a lesser extent by Azapa and Chinchorro. Population, Women Population, and Infant Population exhibit a similar spatial distribution pattern, suggesting a possible correlation. Conversely, the Elderly Population tends to concentrate in the city center, indicating that this sensitivity indicator is less exposed to extreme temperatures.

For the Educational Level indicator, vulnerability is prevalent across a large part of the city, with all census districts showing areas of medium to high vulnerability. Pedro Blanquier, Cancha Rayada, and San José exhibit the most extreme values. Regarding the Employability indicator, the southern part of the city appears to be less vulnerable than the north. Among the indicators that comprise Adaptive Capacity, the central zone and Azapa show lower values for housing quality. Access to water is generally adequate across most of

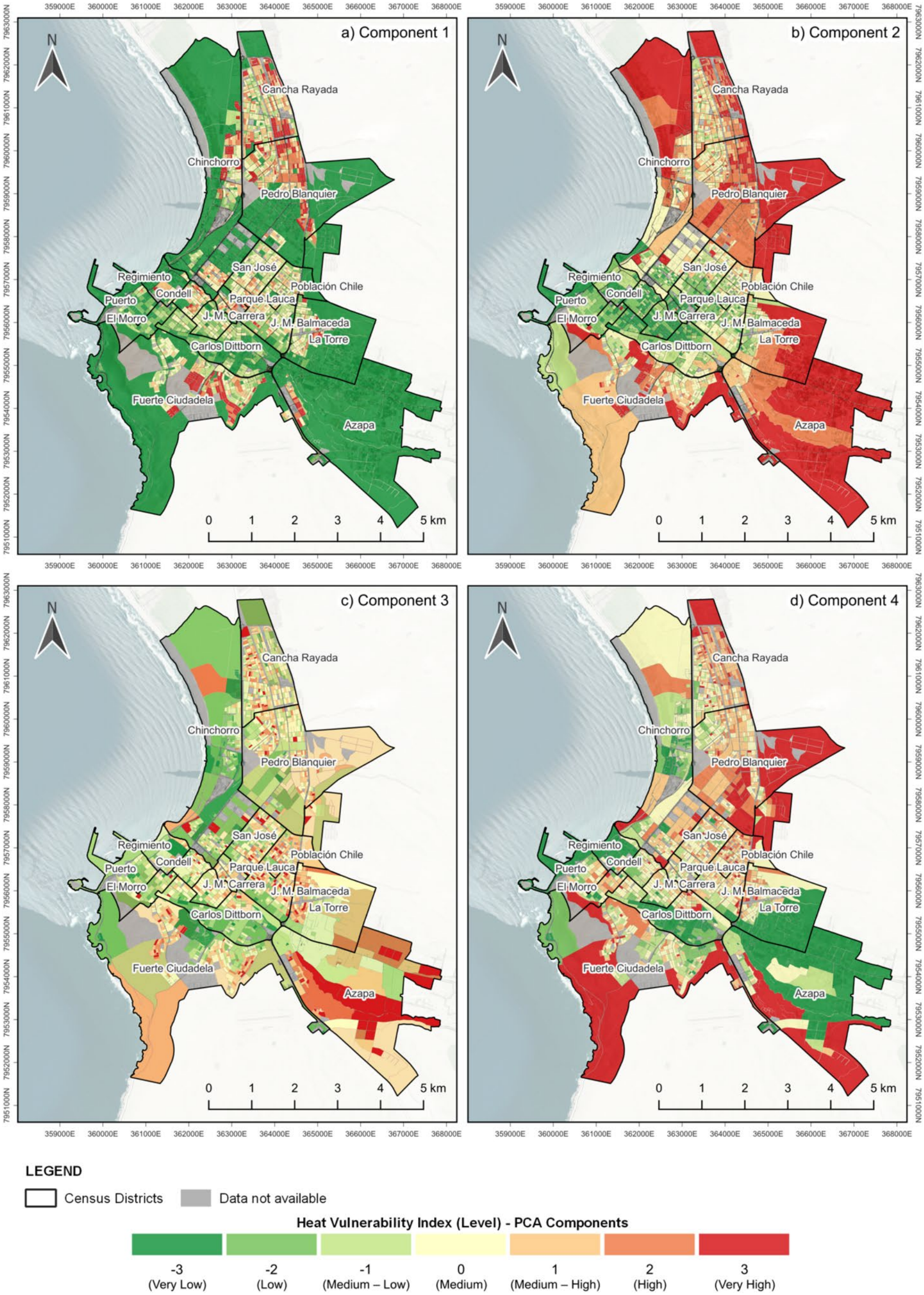

Fig. 4 Spatial distribution of the PCA components (**a**) 1, **b**) 2, **c**) 3, and **d**) 4) of heat vulnerability in Arica

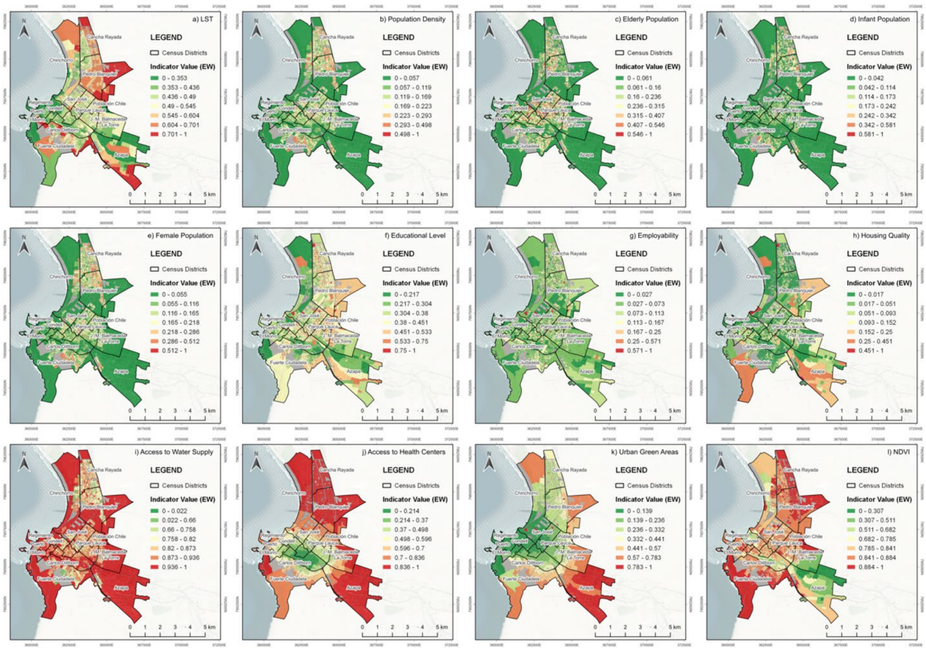

Fig. 5 Spatial distribution of the heat vulnerability indicators in Arica using EW: (**a**) LST, (**b**) population density, (**c**) elderly population, (**d**) infant population, (**e**) female population, (**f**) educational level, (**g**) employability, (**h**) housing quality, (**i**) access to water supply, (**j**) access to health centers, **k**) urban green areas, and **l**) NDVI

the city, with somewhat lesser availability in the districts of Pedro Blanquier and La Torre, while Azapa exhibits significantly limited access. In terms of access to health centers, there is a clear spatial distinction. The districts of Condell, J.M. Carrera, and Carlos Dittborn exhibit less segregation, whereas districts further from the center are particularly vulnerable in this respect. The central and coastal areas benefit from green spaces such as parks and plazas. Although Azapa shows vulnerability in this aspect, it stands out positively in the NDVI indicator, indicating a higher density of vegetation. Pedro Blanquier, Cancha Rayada, and Fuerte Ciudadela continue to demonstrate vulnerability, underscoring the need to increase the presence of green spaces and vegetation in these areas.

3.4 Comparing Spatial patterns among the heat vulnerability indices in Arica

At a general spatial level, both the PCA and EW approaches indicate that the northeast of Arica, specifically in the districts of Cancha Rayada and Pedro Blanquier, exhibits greater vulnerability to heat compared to the rest of the city. Both methods also concur that the central area is the least vulnerable to heat (Fig. 6). However, despite an apparently homogeneous distribution between the two indices, differences in their spatial patterns are evident.

Figure 7 reveals significant disparities between the indices in the Azapa census district, where the PCA method characterizes one of its census tracts as highly vulnerable, while the EW method classifies it as having medium vulnerability. Similarly, in the districts of Chinchorro, Pedro Blanquier, and Puerto, variations occur wherein individual census tracts

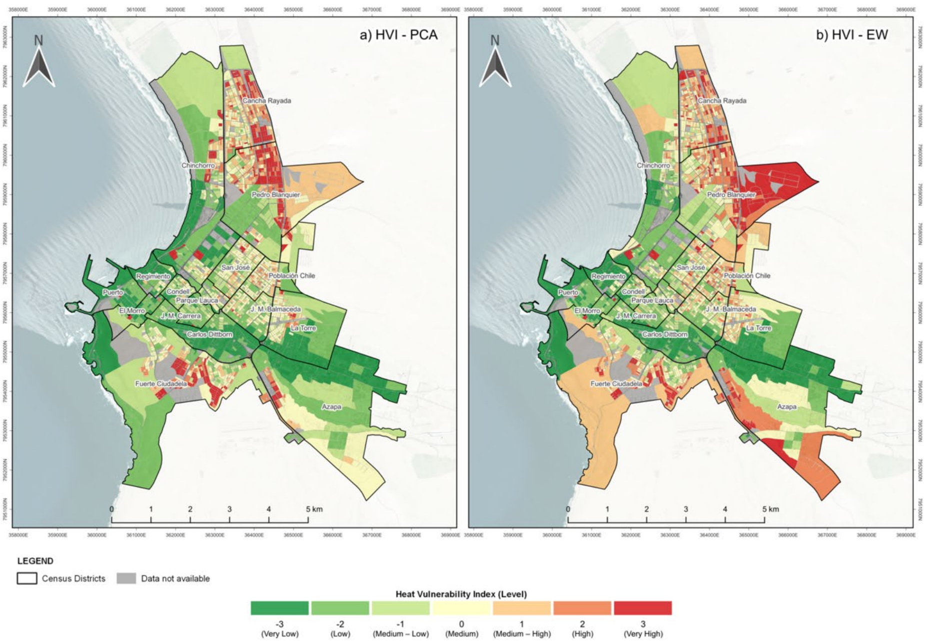

Fig. 6 HVI with (**a**) PCA, and (**b**) EW

range from medium to low vulnerability across the indices. These discrepancies in the general spatial pattern are primarily concentrated in the central zone of the city and peripheral areas, while the trends in the north and south show more consistency between the results.

3.5 LST distribution according to LCZ

LST vary depending the different classes of LCZ (Fig. 8). They are ranked below in descending order: class C with an average of 31.2 °C, class F with 31.0 °C, class 7 with 30.7 °C, class 2 with 30.4 °C, class 8 with 30.1 °C, class E with 30.0 °C, class 3 with 29.7 °C, class 9 with 29.1 °C, class 6 with 29.0 °C, class 5 with 28.7 °C, class 4 with 28.4 °C, class D with 28.1 °C, and class 10 with 28.0 °C.

Compact LCZ classes (2 and 3) show higher LST compared to open classes (4, 5, and 6), meaning that urban areas with higher building density and a larger proportion of impermeable surfaces experience higher surface temperatures.

In particular, the results show that two environmental-type LCZ, C and F stand out for having the highest surface temperatures. Although C is a vegetation zone, its high temperatures can be attributed to direct exposure to the sun and the scarcity of shade characteristic of shrublands in arid environments. On the other hand, F, representing bare or sandy soil, tends to absorb and retain more solar heat, resulting in higher surface temperatures.

Furthermore, the results show that class 7 has the highest LST among urban-type LCZ, with a maximum of 31.9 °C. This observation suggests that this class, characterized by low-rise buildings and a more dispersed distribution, may experience a higher degree of urban warming. Low-rise buildings, being less exposed to shade and wind than taller structures,

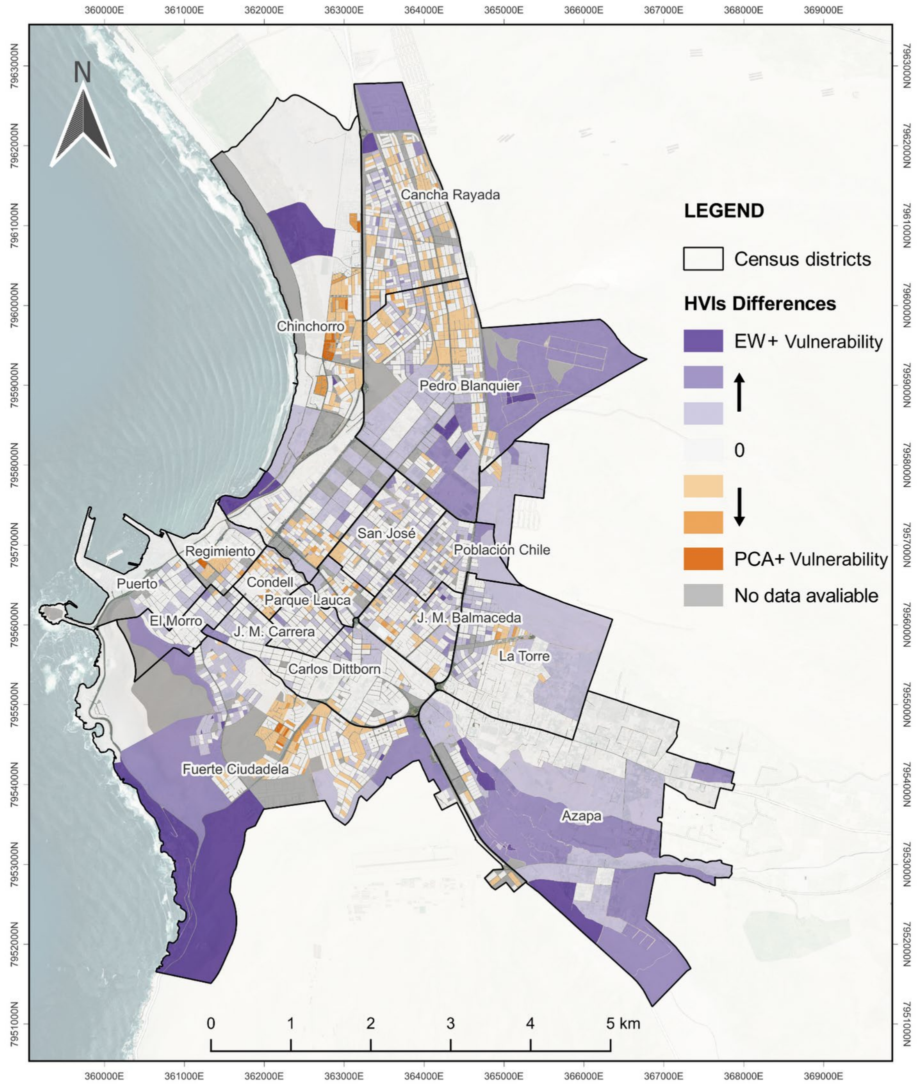

Fig. 7 Differences between HVI

may retain more heat during the day, thereby raising surface temperatures. Additionally, the "lightweight" nature of constructions in this class may influence their ability to accumulate and release heat. Materials used in low-rise buildings, such as concrete and brick, may retain heat more efficiently than taller structures, contributing to the increase in surface temperatures.

However, it is crucial to recognize that within each class, there is significant variability in the observed temperatures. This finding highlights the importance of considering dispersion measures such as the median and interquartile ranges to obtain a more comprehensive understanding of temperature distribution within each class. This variability may be influ-

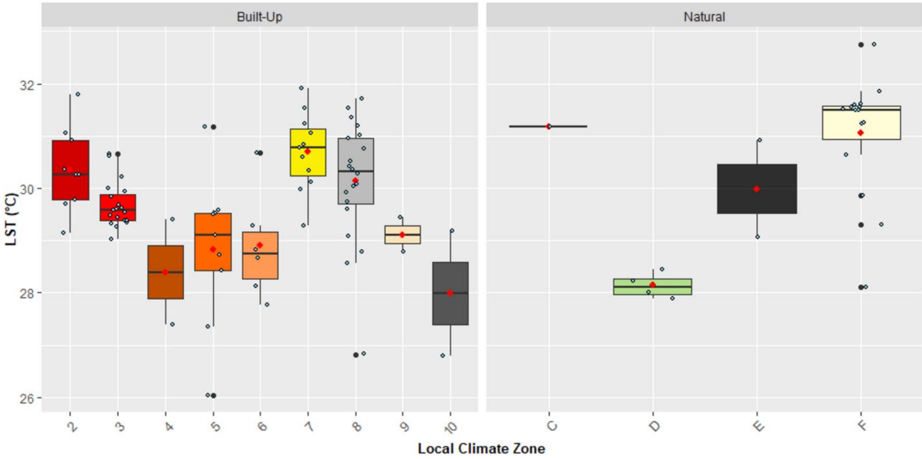

Fig. 8 Boxplot of the LST in the different LCZ in Arica

enced by various local factors, such as building orientation, presence of green areas, and sun exposure, which can modulate the effect of building density and impermeabilization on LST.

3.6 HVI on LCZ

HVI were assessed using the different methodologies: PCA and EW for the different LCZ. The results obtained and a detailed comparison between both methods are presented below, highlighting the zones with higher and lower heat vulnerability, as well as the coincidences and differences between the results (Fig. 9). When analyzing heat vulnerability using the PCA method, it was found that the LCZ classes with the highest maximum values, indicating greater vulnerability, were 8 with a maximum value of 2.285, followed by 2 with a maximum value of 1.554, and 3 with a maximum value of 1.192. These values indicate that, according to PCA, these zones exhibit the highest levels of exposure and susceptibility to extreme heat, being the most critical in terms of vulnerability. On the other hand, analysis using the EW method revealed that the LCZ classes with the highest maximum values were 7 with a maximum value of 1.703, followed by F with a maximum value of 3.431, and 2 with a maximum value of 1.483. Although there is a coincidence in identifying 2 as one of the most vulnerable zones, the EW method highlights 7 as the class with the highest vulnerability, which differs significantly from the results obtained through PCA.

Regarding the zones with lower heat vulnerability, using the PCA method, C class was identified with a constant value of -0.953, followed by 4 with a minimum value of -2.981, and 6 with a minimum value of -2.562. These LCZ exhibited the lowest levels of vulnerability, suggesting lower exposure and susceptibility to extreme heat. Similarly, the EW method identified 4 with a minimum value of -3.567, followed by 5 with a minimum value of -2.941, and 6 with a minimum value of -2.838, as the zones with lower heat vulnerability. These results are consistent with those obtained through PCA, indicating agreement in identifying the least vulnerable zones (Fig. 10).

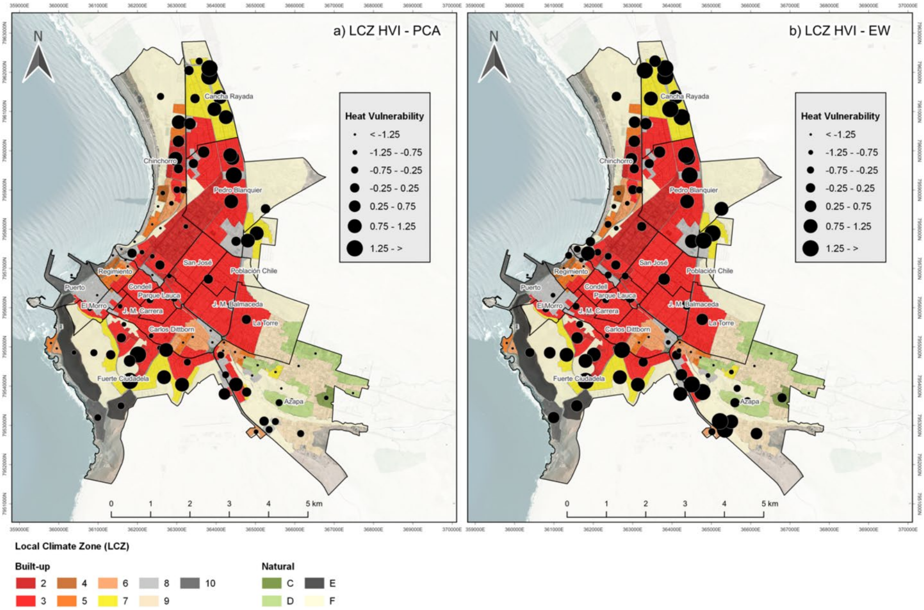

Fig. 9 Heat Vulnerability level from PCA and EW in different LCZ in Arica for (**a**) PCA, and (**b**) EW

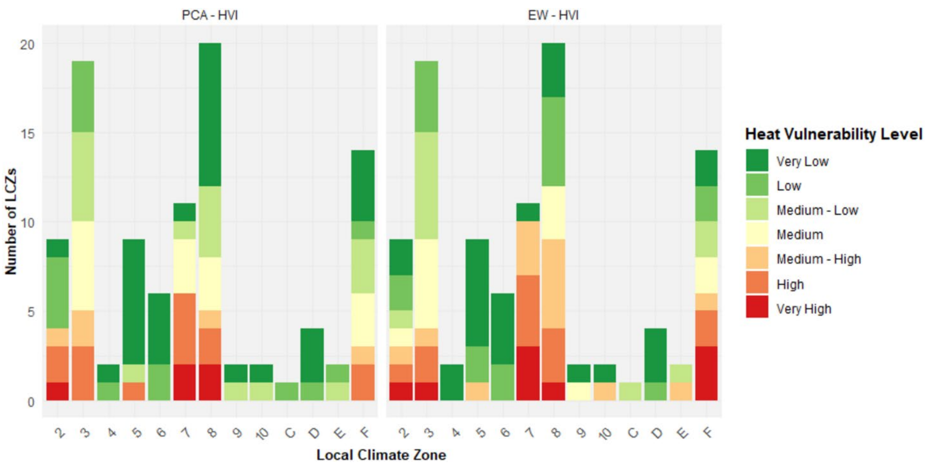

Fig. 10 Differences in heat vulnerability indices in the different local climatic zones of Arica

In summary, although both PCA and EW methods coincide in some of the most and least heat-vulnerable zones, there are significant differences in the magnitude of values and in some of the zones identified as the most vulnerable. The PCA method highlights vulnerability in dense urban areas such as 8 and 2, while the EW method identifies extremely high vulnerability in 7 areas. These differences underscore the importance of considering multiple methodologies for a comprehensive and accurate assessment of heat vulnerability in different classes of LCZ.

4 Discussion

4.1 Exposure and heat vulnerability in the City of Arica

The calculation and mapping of the LST in Arica align with the research conducted by Smith et al. (2021). The findings highlight that the lowest temperatures in Arica are predominantly concentrated along the coastal edge, suggesting a moderate influence of the sea on local thermal regulation. However, when these results are compared to other studies, where the maximum temperature exceeds 36 °C (Al-Ruzouq 2022), the influence is much more significant, particularly in coastal cities affected by cold currents like Arica. It appears that the prevalence of intermediate temperatures in the city and their uniform distribution could be associated with the proximity of urban areas and the pattern of land use.

Regarding high temperatures, which pose significant concerns for population vulnerability, their maximum peaks are found in specific census districts characterized by large areas of arid and unoccupied land. Here, solar radiation impacts directly without absorption by structures or pavement. This observation aligns with the conclusions of Quintana-Talvac et al. (2021), who identified a correlation in Arica between urban heat islands and low socioeconomic levels (e.g., Fuerte Ciudadela, Pedro Blanquier, Cancha Rayada), where the most significant heat is found outside urban morphology in wasteland areas devoid of urban coverage.

Additionally, the Azapa census district presents an intriguing discrepancy between vulnerability indices. Despite high NDVI values, indicating substantial vegetation, it shows high vulnerability in Urban Green Areas and Access to Water Supply. It is essential to note that Azapa is located in a valley of the same name, an arid area where water has been crucial not only for agriculture but also as the primary source of domestic water in Arica (Meseguer-Ruiz and Paillacán 2019).

In that sense, it is relevant to mention that in Chile a HVI has been conducted in the city of Santiago using only the PCA approach (Inostroza et al. 2016). In that work, a sensitivity analysis was carried out that highlighted the crucial impact on indicators such as the Access to Water Supply. The implementation of a similar sensitivity analysis in our study could bring clarity to the understanding of the contrasts between indices according to the most heterogeneous indicators among them.

What has been discussed so far has served the main purpose of this research, to unravel the complexities of heat vulnerability in Arica. In this sense, Romero et al. (2010) encompass the ideas discussed by specifying that the resulting spatial configurations of urban climates are generated from inadvertent and intentional transformations introduced on the regional and local base climates in areas where solar radiation is relatively homogeneous throughout the year, in addition to maintaining in this case a constant influence of the Pacific Ocean.

4.2 Comparison between PCA and EW

The EW method assumes that all indicators equally influence the final composition of indices. In contrast, PCA reduces dimensionality among indicators, and although less intuitive, it provides a nuanced understanding of underlying patterns (Li et al. 2022). Comparisons with other studies, like Tate (2012), have demonstrated the PCA method's effectiveness at various scales and indicators in social vulnerability indices.

The discrepancies identified between the PCA and EW approaches can be attributed to the inclusion of extensive non-residential areas exhibiting elevated values in variables such as LST, NDVI, or service accessibility. Within the EW framework, these peripheral zones—characterized by low residential density—diminish the relative influence of population-based indicators, thereby obscuring significant variations. Conversely, the PCA method more clearly delineates these differences, as the first principal component captures the predominance of demographic variables in the assessment of vulnerability.

Similarly, Guo et al. (2019) also conducted a comparative analysis of the heat vulnerability index in Beijing City, China, using the same approaches. The results of this study pointed out that the spatial pattern of heat vulnerability using PCA tends to produce higher vulnerability in the city center and lower vulnerability in the outer areas of the city compared to the EW method same as this one. However, EW there reveled a north-south, while in this analysis, those areas exhibited were high vulnerability.

Likewise, Liu et al. (2020) conducted a similar investigation. In this case, the results indicated that the EW-focused index outperformed its counterpart. This conclusion was based on the correlation of the models with heat-related mortality data. Regarding spatial patterns, the results of this study were opposite by using PCA underestimate vulnerability in the urban center and overestimate it in suburban areas. However, in our results, the overall geographic distribution of heat vulnerability does not change substantially compared to the works mentioned in this section.

To better understand these methods and the differences in their heat vulnerability values consequently reflected in their spatial patterns, Li et al. (2022) provides an in-depth analysis of their use in constructing HVIs trough a systematic review, exploring the results that can be derived and interpreted from them. Regarding PCA, it remains unclear whether the components formed by the various indicators in the HVI represent statistical relationships that can be explicitly reflected in the different dimensions of vulnerability. This ambiguity can lead to confusion in the analytical logic, as there may be a mismatch between the components and their corresponding indicators, making the practical meaning of these components unclear. As for the EW method, it is often criticized for lacking theoretical foundations, which undermines its credibility. This method is seen as more subjective, as it places the entire decision-making process in the hands of the researcher's judgment.

In this context, to achieve greater clarity regarding the discrepancy between PCA and EW, Cheng et al. (2021) has previously mentioned in another systematic review that it would be helpful for studies related to heat vulnerability to conduct more sensitivity analyses and compare various weighting schemes, which would allow for a more precise clarification of the relative strengths and biases of each method.

4.3 Heat vulnerability in the different LCZ

The results of this study reveal interesting patterns regarding LST and heat vulnerability in different LCZ in Arica. The implications of these findings and their relevance for understanding and addressing heat-related challenges in urban and environmental settings are discussed here.

The assessment of heat vulnerability through PCA and the EW method reveals differences and similarities in identifying the most and least vulnerable local climate zones. While PCA highlights vulnerability in only three urban climate zones such as 2, 8, and 7, the EW method identifies those classes and 3 and F. However, both methods identify class 7 as the most vulnerable to heat in the city of Arica. Moreover, the stratification by LCZ appears to further attenuate this effect under the EW method, likely due to the aggregation of more homogeneous spatial units and a clearer distinction between built-up and non-built-up areas. Similar results have been found elsewhere in the world with cities of different environmental types (Zhou et al. 2021). The income profile of residents and the accessibility of social infrastructure in compact development zones are the reasons for this finding (Iqbal et al. 2023), as low-income tower block communities are more vulnerable to heat (Bu et al. 2024).

It is crucial to recognize the significant variability in observed temperatures within each LCZ class. This variability may be influenced by local factors such as building orientation, the presence of green areas, and sun exposure. This variability underscores the need to consider dispersion measures and local factors when interpreting results and developing heat mitigation and optimization strategies (Zhang et al. 2023).

4.4 From results to urban strategy recommendations

With both approaches moderately heterogeneous, there are census blocks to pay attention to, as the health of their populations may be compromised. In Madrid, Sanchez et al. (2017) determined that vulnerable populations are located in warmer areas, so people living in these areas are more prone to health problems, especially at-risk groups that are exposed to extreme heat events.

According to the results, in Arica adults are relatively more protected than the child population in areas where education in the populations is low and access to health is extremely complicated. At the level of dimensions, although the exposure is not extreme, the adaptive capacity is alarming and it would be ideal to have more information on Sensitivity variables.

In that sense, while a scarcity of green areas has been observed in Arica, the literature suggests that these areas, along with other factors, can help mitigate the negative effects of heat exposure, especially those related to urban morphology, such as building distribution and construction materials (Gourfi et al. 2022). Therefore, it is important not to overlook those factors more associated with the geographical condition of an arid city like Arica in order to address urban planning more appropriately and effectively determine mitigation practices in reducing heat vulnerability.

4.5 Limitations of the study

The study's limitations include the lack of model validation, which some researchers have addressed by correlating indices with mortality data (Mallen et al. 2019; Liu et al. 2020) or LST (Mushore et al. 2018). In Chile, accessing health data can be challenging due to its sensitive nature, which might stigmatize certain populations, particularly given the study's scale. Moreover, the indices that have been developed in this study represent a combination of several indicators covering various aspects related to heat vulnerability composed of the dimensions of Exposure, Sensitivity and Adaptive Capacity. Consequently, attempting to correlate this index with a single indicator, such as LST, may be meaningless and may not provide a complete understanding of the complex underlying relationship. In other words, seeking a correlation with a single factor about the index could minimize the richness of the interaction between these components.

5 Conclusions

The objective of this study was to evaluate heat vulnerability in Arica using two distinct methodological approaches. The results showed that both indices identified Pedro Blanquier and Cancha Rayada as the census districts most susceptible to heat, particularly in the northern part of the city. However, discrepancies between the indices were noted. The PCA approach emphasized vulnerability in the central zone, while the EW approach highlighted heat vulnerability around the central zone and in the southwest of the city.

The primary difference between the two methods lies in how indicators are weighted. The EW method distributes weight uniformly across all indicators, whereas PCA assigns varying weights during the generation of its components. It is crucial to acknowledge that the mapping outcomes of these methods can be significantly influenced by the choice of indicators and the scale of the study.

However, the choice of mapping the indices using census districts offers important support for comparison between sectors and decision-making aimed at mitigating heat vulnerability in Arica. At the same time, considering the subdivisions of these districts, the census blocks enable the practical operability of mitigation strategies, which facilitates effective management of the efforts involved.

Therefore, it is not only advisable to pay attention to those areas that show a coinciding vulnerability in both approaches, but also to those with low levels of vulnerability. This will allow the authorities address critical areas but also to understand and preserve less vulnerable spaces, which will enrich the strategies for the most affected districts.

Furthermore, it is important to highlight that LCZ play a fundamental role in the distribution of heat vulnerability in Arica. The variability of environmental and urban conditions within each LCZ class influences the exposure and susceptibility of different areas to extreme heat. Therefore, understanding how these zones affect the spatial distribution of heat vulnerability is crucial for developing effective adaptation and mitigation strategies in the city.

Finally, this study showed preliminary results that pose new challenges for future research, such as improving the representativeness of the indicators and trying to collect data that could allow analyzing the fidelity of the models, as with data on heat-related diseases. In addition, it is suggested to explore other forms of indexing as well as to consider different weighting methods in search of a more representative model to address and understand heat vulnerability in Arica and other arid or coastal desert cities in Chile and the world.

Acknowledgements Angelo Olivares thanks the financial support provided by the Chilean National Agency for Research and Development (ANID) through the 2022 Master's Abroad Scholarship (Application No. 73220349). Pamela Smith and Pablo Sarricolea thank the Center for Climate and Resilience Research (CR2, CONICYT/FONDAP/15110009). Oliver Meseguer-Ruiz thanks the UTA Mayor project 5831-24.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Angelo Olivares and Oliver Meseguer-Ruiz. The first draft of the manuscript was written by Angelo Olivares and Oliver Meseguer-Ruiz. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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